Assignment Deadline Optimizer

AI-powered personal deadline tracker that analyzes workload, prioritizes tasks, and tells you exactly what to work on and when

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Context

End User

University students managing 4–5 courses per semester, each with multiple assignments, labs, and exams with overlapping deadlines.

Problem

Students face 4–5 overlapping deadlines every week, don't know where to start, panic, and end up submitting late or producing low-quality work.

Product Idea

AI-powered personal deadline tracker that analyzes your workload, prioritizes tasks by urgency and complexity, and tells you *exactly what to work on and when*.

FastAPI Backend

PostgreSQL Database

React SPA Frontend

MCP Agent Nanobot

Implementation

Version 1 – Core Feature

- Priority scoring algorithm: *urgency + weight + status*
- AI recommendations per task
- Risk levels: low / medium / high / critical
- Conflict detection for overlapping deadlines
- Progress tracking per course

Version 2 – Improvements

- Full CRUD via web UI (create/delete deadlines)
- Management panel with forms
- Dashboard with charts (Chart.js)
- MCP server – 7 tools for AI agent
- Docker Compose – 8 services
- OpenTelemetry observability

TA Feedback Addressed

Added management panel for CRUD operations; improved conflict warnings visibility in UI; added productivity statistics endpoint; created comprehensive test suite (58 tests passing).

Tech Stack: `FastAPI` `PostgreSQL` `React+TS` `MCP` `Docker`

Demo

Pre-recorded video demonstration of Version 2 with voice-over ($\leq$ 2 minutes)

Deadline Optimizer | Tasks Dashboard Manage

Algorithm Quiz | CS101 | Score: **120** | Due tomorrow

Derivatives Homework | MATH201 | Score: **70** | Due in 2 days

Mechanics Lab Report | PHYS301 | Score: **50** | Due in 4 days

5	12	37	7
Courses	Assignments	Unit Tests	MCP Tools

Video file: https://drive.google.com/file/d/1fXM1kWkiodFm7xbiYehvISY3W8bfFa-/view?usp=

Links

GitHub Repository

github.com/Rena-ln/
se-toolkit-lab-9

Deployed Product

http://10.93.24.215:8080/

API Key: dev-api-key-change-in-production

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